

- This exam contains 10 pages (including the front and back of this cover page) and 19 problems. Check to see if any pages are missing.
- This is the third exam and will count for 15% of your final grade. You have from the end of class on Thursday, June 7th until 12:00 PM (noon) on Friday, June 8th to complete this exam. When you are finished, return your completed exam to Ross Hall, office 303 by placing it in the envelope attached to the door.
- You may use one index card of hand-written notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- If a question is followed by work, then work of some form is required. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. A correct answer that has no supporting work when work is required will not receive full credit.
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 150 points and 20 bonus points total in this exam. No "free" points will be included in your score.
- If a question has multiple parts, the points for that question are divided evenly among the parts.
- Do not write in the $\checkmark$ + - box, because this is for grading.
The grading scale (except True/False) is: $\checkmark$ = 100%, + = 50%, - = 0%

Page:	3	4	5	6	7	8	9	Total
Points:	15	25	30	20	20	20	20	150
Score:								

1. (5 points) While shopping for clothes, you find a shirt that you like on sale. The price of a shirt is marked down by 30%, and you also have a coupon that allows you to take another 20% off of the sale price. If you purchased the shirt for \$10, then what was the original price prior to the 30% mark-down?

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- (a) \$14.67
(b) \$16.86
(c) \$17.86
(d) \$166.67
(e) None of the above.

2. (5 points) Your university charges an annual room and board fee of \$4,200. You choose the deluxe meal program, so your room and board fee is increased by 15%. What will be your increased room and board fee? Give your answer as a decimal rounded to the nearest hundredth.

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- (a) \$4626.00
(b) \$4750.00
(c) \$4830.00
(d) \$5052.00
(e) None of the above.

3. (5 points) The local Rent-to-Own Store offers you the following plan to purchase a bedroom set. The bedroom room set retails for \$2,350, but you will have two years to pay it off using a simple interest loan. If you pay back \$2,560, then what was the APR on this loan? Give your answer as a percentage rounded to the nearest tenth of a percent.

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- (a) 4.5%
(b) 4.9%
(c) 5.2%
(d) 5.6%
(e) None of the above.

4. (5 points) After some questionable (though good-spirited) antics during your college's annual homecoming party, your friends need to bail you out of jail. A bail bondsman gives your friends a \$5,000 simple interest loan at 15.25% APR. After six months, you go to pay the loan. How much do you owe? Give your answer as a decimal rounded to the nearest hundredth.

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- (a) \$5123.50
- (b) \$5381.25
- (c) \$8932.75
- (d) \$9575.00
- (e) none of the above

5. (10 points) After several sleepless nights of on-line poker youve made quite a bit of money. You deposit the money into a brand new savings account which offers an annual percentage rate of 5.75% compounded weekly. You forget about the account and five years later find that it contains \$20,200. How much money did you originally deposit into the account? Give your answer as a decimal rounded to the nearest hundredth.

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- (a) \$14,252.23
- (b) \$15,252.15
- (c) \$18,214.15
- (d) \$15,155.17
- (e) None of the above.

6. (10 points) Federal Savings Bank is offering a 6-month CD (certificate of deposit) with an annual percentage rate of 4.5% compounded daily. What is the APY (annual percentage yield) on the CD? Give your answer as a percentage rounded to the nearest tenth of a percent.

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- (a) 4.7%
- (b) 4.8%
- (c) 4.9%
- (d) 5.0%
- (e) None of the above.

7. (10 points) At the end of each month you deposit \$250 into an account which earns an annual percentage rate of 6% compounded monthly. How much money will you have in the account after 40 years? Give your answer as a decimal rounded to the nearest hundredth.

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- (a) \$120,000.00
(b) \$315,473.28
(c) \$497,872.68
(d) \$589,545.27
(e) None of the above.

8. (10 points) Your university charges an annual tuition of \$8,200. If you agree to take part in a work-study program, your annual tuition is only \$7,600. What is the percentage decrease in the tuition if you take part in the work-study program? Give your answer as a percentage rounded to the nearest tenth of a percent.

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9. (10 points) When you were born, your grandmother purchased a low-risk government savings bond for \$4350. The bond pays 3.25% annual simple interest. Determine the future value of the bond after seven years. Give your answer as a decimal rounded to the nearest hundredth.

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10. (10 points) You have recently come across a rather large amount of money. Because you have just finished the financial math lectures in your college math class, you decide that you should invest the money in a long-term bond. However, your friend, Dale, has recently asked for a loan as well. Given that you can only afford one of the following choices, which one nets you a large return?
- (a) **Federal Treasury Bond:** 10 year, \$15,000 bond (which pays simple interest) with an APR of 4.44%.
- (b) **Dale's Loan:** 7 year, simple interest loan of \$11,000 with an APR of 7.43%.

11. (10 points) While shopping, you find an item whose price is marked down by 50%. Even better, you were lucky enough to have a coupon that allows you to take an additional 20% off of the sale price. If you purchased the item for \$100, then what was the original, pre-sale price of the item? Give your answer as a decimal rounded to the nearest hundredth.

12. (10 points) Let r , the interest rate of a loan for a given time period (i.e. something like the annual percentage rate) be a percentage expressed in decimal form. Further, let P be the principal, let t be the term of the loan, and let F be the final amount the borrower will repay. State the simple interest formula in these terms and describe – intuitively, with words – what this formula tells you.

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13. (10 points) After selling a bunch of your old junk on e-bay last year, you earned \$5,800. You spent the money on a fishing boat, but after the first three years of owning the boat, you realized that you don't actually like fishing. You decide to sell the boat on craigslist, and you calculate that even the best offer you got amounts to losing 15.3% of the money you initially spent. Assuming the boat decreased in value by a constant amount each year (i.e. a constant percent of its initial purchase price), what percent of the initial purchase price did you lose each year by owning this boat? Give your answer as a decimal rounded to the nearest hundredth.

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14. (10 points) You have recently come across a rather large amount of money. Because you have just finished the financial math lectures in your college math class, you decide that you should invest the money in a long-term CD (certificate of deposit). The financial section of your local paper has the following advertisements:
- (a) Countryside Credit Union: 10 year CD with an APR of 5.44% compounded monthly.
 - (b) Northeast Bank and Trust: 10 year CD with an APR of 5.43% compounded continuously.

Which institution is offering a better deal? Explain.

15. (10 points) You purchase a condominium after graduating from college. Each year, the assessed value of the condominium increases by 5%. After five years, what is the total percentage increase on the assessed value of the condominium? Give your answer as a percentage rounded to the nearest tenth of a percent.

16. (10 points) You are considering two savings accounts to put \$600 in for 10 months. Account A pays 39.72% simple interest. Account B pays $x\%$ APR compounded quarterly. Solve for x so that the same amount of money would be in either account at the end of 10 months.
- (HINT: "Quarterly" means "every three months")

17. (10 points) At the world puzzle fair, the largest puzzle ever was assembled by a team of over 100 people. However, as the puzzle neared completion, the participants estimated that this 1,800,000 piece puzzle was missing about 11% of its pieces. About how many pieces did they assemble?

18. (10 points (bonus)) A savings account that pays 6% APR compounded monthly is equivalent to what percent APR compounded quarterly?

(HINT: Set up the appropriate form of $F = P(1 + r)^t$ for each case. Then, set the F 's equal, solve for your missing r , and convert your solution into an APR)

19. (10 points (bonus)) You purchase a new car after graduating from college for $\$P$. Each year the assessed value of the car decreases by 8.5%. After five years, what is the total percentage decrease on the value of the car? Give your answer as a percentage rounded to the nearest tenth of a percent.